Krishna Durgavarjhula

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EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY August 2026 - May 2030 (expected)
Bachelor of Arts in Computer Science
Current/Enrolled Relevant Coursework: CS 61A, Math 52, CS 61B, Math 53, Physics 7A
AMERICAN HIGH SCHOOL August 2022 - May 2026
High School Diploma, GPA: 3.74 (unweighted)
Relevant Coursework: AP Computer Science A (AP CSA), AP Physics 1, AP Calculus BC, AP Physics C (Mechanics and E&M), AP Government, AP Economics, AP Calculus BC, Entrepreneurship, Digital Electronics
Clubs: Computer Science Club (Secretary & Instructor), Quiz Bowl (Division C Team Member)
Test Scores: AP CSA: 5; AP Physics 1: 4; AP U.S. History: 4; AP CSP: 4; AP Calc BC: 4; AP Macroeconomics: 5; SAT: 1510
Other Relevant Coursework: Intro to C++ Programming, 2D Design, AP Computer Science Principles (APCSP)

SKILLS

- **Concepts:** Object-Oriented Programming, Design Principles, Autonomous Systems, Project Management, Debugging and Testing, Data Structures and Algorithms, Full-Stack Development
- **Languages**: Java, Python, C++, HTML/CSS/JavaScript, MATLAB
- **Databases/Frameworks/Libraries/Tools**: Node.js, NumPy, TensorFlow, Keras, React, Claude Code, Codex, Antigravity

PUBLICATIONS

Arora, J.S., Durgavarjhula, K., & Bansal, R. (2025). An Assistive Sensory-Based System for Integrated Fall and Obstacle Detection. 2025 3rd International Conference on Computer System, Information Technology, and Electrical Engineering (COSITE), 37-42.

WORK EXPERIENCE

STEALTH STARTUP | *Intern* July 2026 - Present
- Expanded core validation logic and backend services using TypeScript to increase rule coverage and ensure system performance.
- Applied automated testing suites, GitHub, and continuous integration workflows to validate changes and streamline development.

LIFT AND SHIFT FOUNDATION | *Web Design Intern* January 2024 - June 2026
- Designed and developed interactive UI features on the Lift and Shift website using HTML, CSS, and Javascript for the Lift and Shift IDE, which teaches block coding and 3D modeling for veterans re-entering the workforce.
- Applied full-stack web development, GitHub, collaborative development, and web architecture practices.

PROJECTS

UC BERKELEY ROAR ACADEMY | *Student* July 2025 - August 2025
- Performed machine learning simulations in Python for autonomous driving using deep neural networks, gradient descent, and reinforcement learning.
- Gained experience with Python libraries, AI/ML, and software simulation of real-world environments, working with UC Berkeley faculty and student researchers.

BREATHE APP | *Developer* July 2024 - August 2024
- Developed an Android app where users can log their respiratory symptoms and view daily weather data to understand their triggers over time.
- Utilized Javascript for native full-stack development, Node to deploy the backend, Firebase for dynamic user authentication, and Expo for iterative testing, and published the app to the Google Play store.

ALAMEDA COUNTY SCIENCE & ENGINEERING FAIR | *Team Member* August 2024 - December 2025
- Worked with a team of three to build an environmentally-friendly smart cane to assist seniors in avoiding obstacles and detecting falls along with a caretaker mobile app.
- Developed a system with ultrasonic sensors, a gyroscope, a Bluetooth Module, and an ergonomic handle design with a pulse sensor, haptic motors, and a thermoelectric generator that uses body heat to partially power the system.
- Developed C++ algorithms for ultrasonic terrain profiling, fall detection, and pulse tracking. Integrated via Bluetooth with the Android app, which displays user location, BPM, and fall alerts.
- Won 1st place category award in ACSEF, qualified for CSEF, and won 1st place team engineering project at STEM4ALL. Currently patent-pending, and presented and published at the IEEE-sponsored COSITE 2025.

BLUESTAMP ENGINEERING | *Student* July 2023 - August 2023
- Designed and built a phone-controlled robotic arm using servos, C++, and Arduino, controlled via a Bluetooth module using an app.
- Gained experience in digital electronics, hardware-software integration, project management, and iterative development.
- Prepared a presentation and demonstrated the project to an audience. Documented my work at kri-dur.github.io.

CODING TOMORROW INITIATIVE | *Instructor* June 2025 - July 2025
- Volunteered as a web development instructor teaching a class to 20 middle schoolers online at a non-profit.
- Implemented a web development curriculum with lesson plans, coding examples, and homework assignments.

AWARDS

Alameda County Science & Engineering Fair (ACSEF) 1st Place Category Award, STEM4ALL 1st Place Engineering Project (Team), California Science & Engineering Fair Certificate, BlueStamp Engineering Certificate Of Excellence, AP Scholar With Distinction, College Board School Recognition Award, National Merit Scholar Letter of Commendation